

Course code	CC-A08			
Category	Core Course			
Course title	Computer System Architecture			
Course ID	241/MCA/CC302			
Scheme and Credits	L	T	P	Credits
	3	1	-	4
Theory Internal	30			
Theory External	70			
Total	100			
Duration of Exam	3 hrs			

Note: The examiner will set nine questions in total. Question one will have seven parts from all units and the marks of first question will be of 20% of total marks of Question Paper and the remaining eight questions to be set by taking two questions from each unit and the marks of each question from Question no. 2 to 9 will be 20% of total marks of Question paper. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OUTCOMES:

CO1. Understand and apply number systems including Binary, Octal, Hexadecimal, Decimal, and perform interconversion of numbers.

CO2. Understand the architecture and components of Basic Computer Design, including Computer Instructions and Types, Instruction Cycle.

CO3. Analyze CPU Design principles, including CPU Registers, Micro-operations and their types.

CO4. Describe the organization and functionality of Input/Output Systems including Peripheral Devices, Input-Output Interface.

UNIT-I

Number System: Number System: Binary, Octal, Hexadecimal, and Decimal; 1's and 2's Complements; Interconversion of Numbers.

Codes: Weighted and Non-weighted Codes, BCD Codes, Gray Codes, Self-complementing Codes, Error-Detecting/Correcting Codes, Alphanumeric Codes, Hamming Codes.

Floating Point Numbers: Binary Arithmetic, Binary Addition and Subtraction, 2's Complement Arithmetic, Booth Coding, Binary Multiplication.

Logic Design: Logic Gates, Truth Tables, Boolean Algebra, Boolean Expressions- Variables and Literals, Equivalent and Complement of Boolean Expressions, Theorems of Boolean Algebra, SOPs & POSs of Boolean Expressions.

UNIT-II

Combinational Circuits: Combinational Logic, Arithmetic Circuits - Adder and Subtractor, BCD Adder, Code Converters, Parity Generators/Checkers, Multiplexers, Demultiplexers, Decoders, Encoders.

Sequential Circuits: Latches, RS Flip-Flop, , JK Flip-Flop, Master-Slave Flip-Flops.

UNIT-III

Basic Computer Design: Computer Instructions and Types, Instruction Set, Instruction Cycle, Instruction Formats, Addressing Modes, Computer Registers, Bus System, Register Transfer Language Terminology.

CPU Design: CPU Registers, Micro-operations and Types, Design of ALU, Control Unit Design - Microprograms, Control Unit of a Basic Computer, Timing and Control: Hardwired and Micro-programd Controlled Unit, Architectures - RISC, CISC, Scalar, Superscalar, and Pipelined Architectures.

UNIT-IV

Input/Output Organization: Peripheral Devices, Input-Output Interface, Asynchronous Data Transfer, Mode of Transfer, Priority Interrupt, Direct Memory Access, Input-Output Processor, Serial Communication.

Advanced Architecture: Introduction to Parallel Processing - Pipelining, Parallel Computer Structures, Architectural Classification, Pipelining & Vector Processing; Instruction and Arithmetic Pipelines, Principles of Designing Pipelined Processors.

Textbooks & References:

1. Mano, M. M. Digital Logic and Computer Design. Prentice-Hall of India.
2. Gill, Nasib Singh and Dixit, J. B. Digital Design and Computer Organisation. University Science Press (Laxmi Publications), New Delhi.
3. Stallings, William. Computer Organisation& Architecture.
4. Anand Kumar. Fundamentals of Digital Circuits. PHI.
5. Hwang, Kai. Advanced Computer Architecture. McGraw-Hill International.
6. Mano, M. M. Computer System Architecture. Prentice-Hall of India.
7. Tokheim. Digital Electronics. TMH.

